Lab 1a – Raw Security Logs, Rule-Based Detection, MITRE ATT&CK and SOC Dashboard

AI-Enhanced Cybersecurity · Individual submission

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Repository used: https://github.com/melofon/HIT-ai-cybersecurity-labs (labs/lab1a-mitre-soc)

1. Docker verification and dashboard screenshot

Docker Desktop 4.96.3 was running (Client and Server present). The detector and dashboard containers were built with docker compose up --build -d and reported Up. The simulator stayed stopped until Activity D.

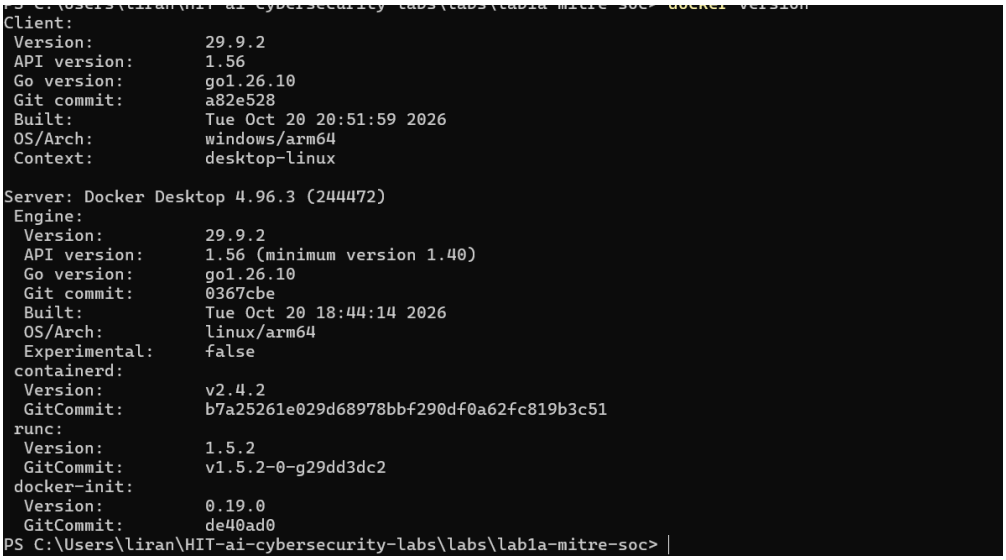


Figure 1. docker version showing Client and Server.

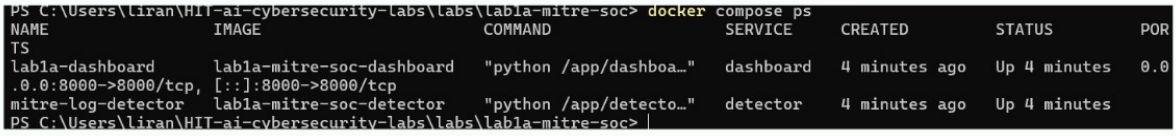


Figure 2. docker compose ps: lab1a-dashboard and mitre-log-detector are Up (port 8000).

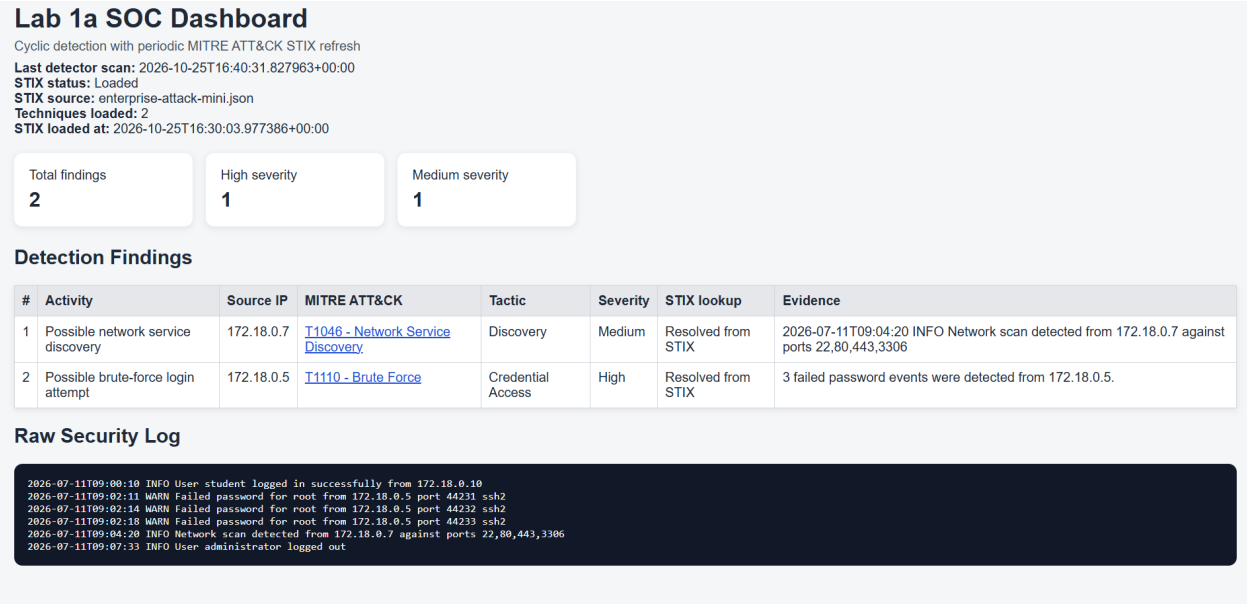


Figure 3. SOC dashboard on the clean baseline log.

2. Initial finding count (clean baseline)

The baseline logs/security.log contained 6 lines (verified with (Get-Content .\logs\security.log).Count). findings.json reported finding_count = 2 (failed_password_threshold 3, scan_interval_seconds 5):

Finding	Source IP	Technique	Severity
Possible brute-force login attempt	172.18.0.5	T1110 – Brute Force	High
Possible network service discovery	172.18.0.7	T1046 – Network Service Discovery	Medium

3. Evidence → rule → technique → tactic → uncertainty

Evidence (raw log)	Detection rule	Technique	Tactic	Uncertainty
Lines 2–4: three "Failed password for root from 172.18.0.5" (ports 44231–44233)	At least 3 failed-password events from the same source IP (detector.py, FAILED_PASSWORD_THRESHOLD = 3; no time window)	T1110 Brute Force (High)	Credential Access	Does not show whether any login succeeded, how fast the attempts happened, or whether they were user typos. Spreading attempts across IPs evades the rule (Experiment C2).
Line 5: "Network scan detected from 172.18.0.7 against ports 22,80,443,3306"	Log line contains the text "network scan detected" (detector.py, line 190)	T1046 Network Service Discovery (Medium)	Discovery	Based on message text only, not on network flows. The source of the message is unknown; it could be an authorized scan or a mis-generated log line (Experiment C4).

A rule match gives a candidate technique, not confirmed adversary intent, identity, compromise or root cause.

4. Experiments C1–C4

Exp.	Changed log lines	Predicted	Actual	Explanation
C1	Deleted the failed-password line with port 44233 (2 failures left for 172.18.0.5).	T1110 disappears; T1046 stays; total 1.	T1110 gone (High 0); T1046 stayed; total 1. Correct.	T1110 needs ≥3 failures per IP; only 2 remained. T1046 depends only on the scan text, which was unchanged.
C2	Restored 3 failure lines from three IPs: 172.18.0.5 (44231), 172.18.0.6 (44232), 172.18.0.8 (44233).	No T1110; T1046 stays; total 1.	No T1110; T1046 stayed; total 1. Correct.	The detector groups failures by source IP; each IP had one failure. Distributed attempts evade this rule.
C3	Appended 3 failures from 192.0.2.55 (ports 40001–40003).	New T1110 for 192.0.2.55; T1046 stays; total 2.	New T1110 for 192.0.2.55 (High); T1046 stayed; total 2. Correct.	One IP reached 3 failures (count ≥ threshold). Older IPs still had 1 each.
C4	Appended "Network scan detected from 192.0.2.77 against ports 22,80,443".	Extra T1046 for 192.0.2.77; total 3.	Extra T1046 (Medium) for 192.0.2.77; total 3 (High 1, Medium 2). Correct.	The line contains the trigger text, so a new candidate finding is created; the rule does not infer scans from raw flows.

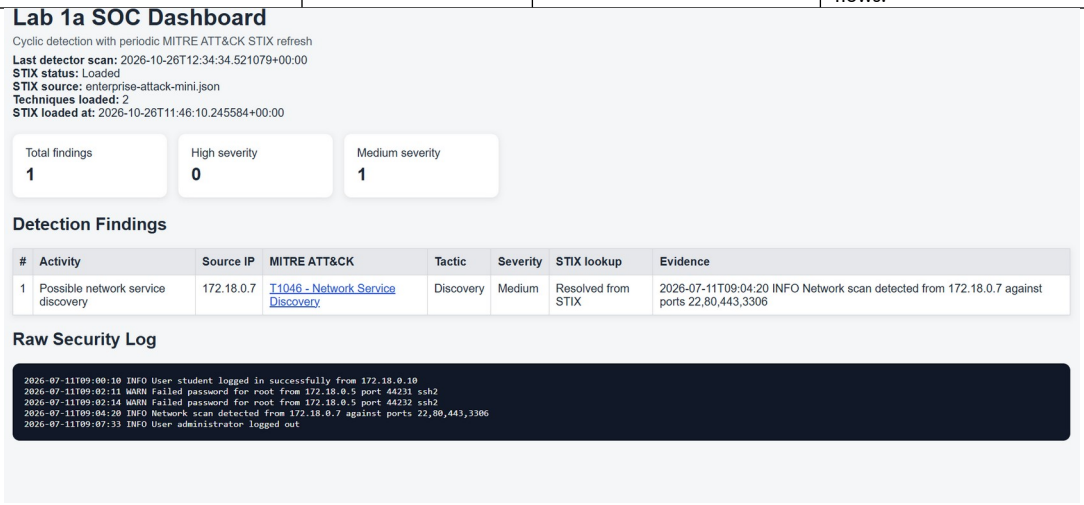


Figure 4. Experiment C1: 1 finding (T1046 only).

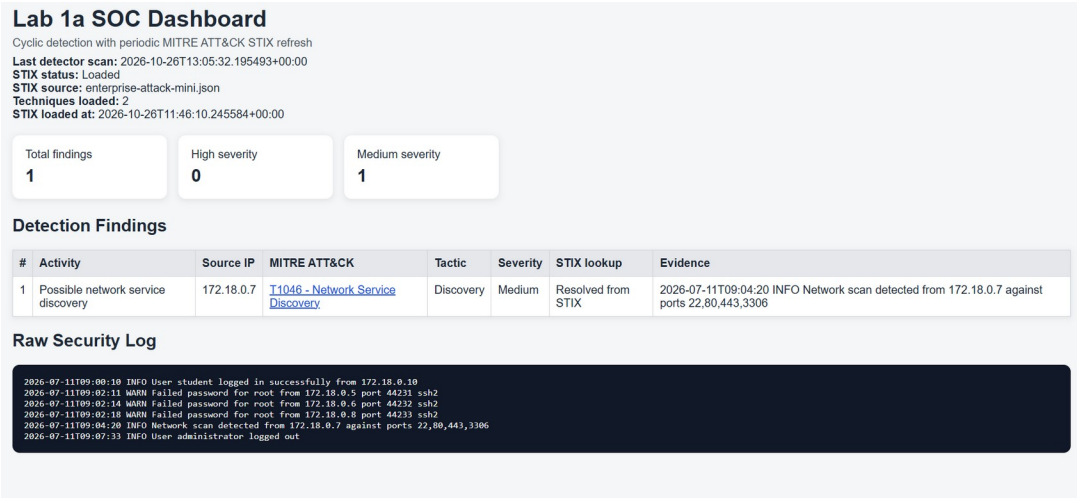


Figure 5. Experiment C2: 1 finding; failures from .5, .6 and .8.

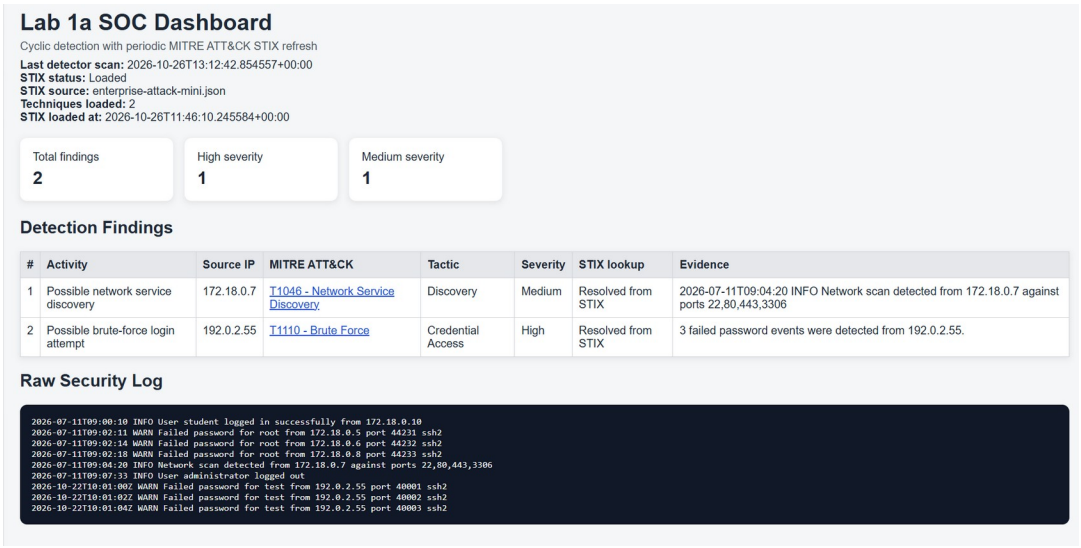


Figure 6. Experiment C3: new T1110 for 192.0.2.55; 2 findings.

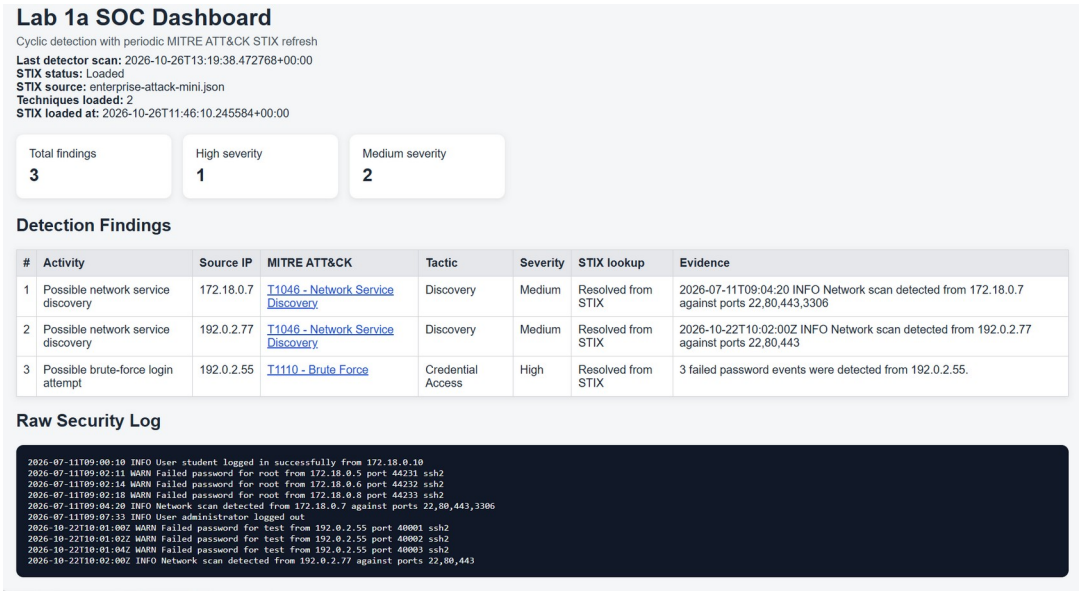
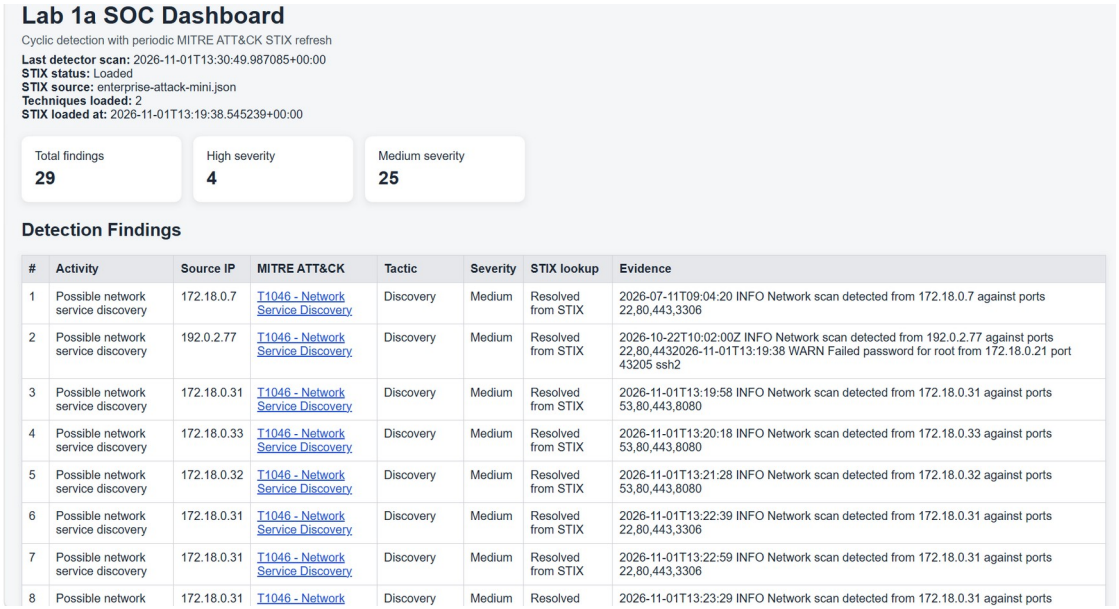


Figure 7. Experiment C4: second T1046 for 192.0.2.77; 3 findings.

5. Simulator observation

With the optional simulator started (docker compose --profile simulation up --build -d), it appended 1 or 3 synthetic events to security.log about every 10 seconds. Findings grew from 3 to 19 and then to 29 (High 4, Medium 25) within a few minutes. New sources were 172.18.0.21, .22 and .23 (failed passwords → T1110, with 17, 12 and 33 failures), and 172.18.0.31, .32 and .33 (scan notices → T1046); benign logins from 172.18.0.10–.12 produced no finding. Counts keep rising because the detector rescans the whole log and counts failures cumulatively with no time window. The run is less reproducible than the manual experiments because event content, IPs and counts are generated at runtime.

The first simulator event was appended directly after my edited last line (the file had no trailing newline), so two events merged into one line and the evidence text of the 192.0.2.77 finding is garbled. This shows that text-based rules are sensitive to log formatting.



Lab 1a SOC Dashboard

Cyclic detection with periodic MITRE ATT&CK STIX refresh

Last detector scan: 2026-11-01T13:30:49.987085+00:00
STIX status: Loaded
STIX source: enterprise-attack-mini.json
Techniques loaded: 2
STIX loaded at: 2026-11-01T13:19:38.545239+00:00

Total findings	High severity	Medium severity
29	4	25

Detection Findings

#	Activity	Source IP	MITRE ATT&CK	Tactic	Severity	STIX lookup	Evidence
1	Possible network service discovery	172.18.0.7	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-07-11T09:04:20 INFO Network scan detected from 172.18.0.7 against ports 22,80,443,3306
2	Possible network service discovery	192.0.2.77	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-10-22T10:02:00Z INFO Network scan detected from 192.0.2.77 against ports 22,80,4432026-11-01T13:19:38 WARN Failed password for root from 172.18.0.21 port 43205 ssh2
3	Possible network service discovery	172.18.0.31	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-11-01T13:19:58 INFO Network scan detected from 172.18.0.31 against ports 53,80,443,8080
4	Possible network service discovery	172.18.0.33	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-11-01T13:20:18 INFO Network scan detected from 172.18.0.33 against ports 53,80,443,8080
5	Possible network service discovery	172.18.0.32	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-11-01T13:21:28 INFO Network scan detected from 172.18.0.32 against ports 53,80,443,8080
6	Possible network service discovery	172.18.0.31	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-11-01T13:22:39 INFO Network scan detected from 172.18.0.31 against ports 22,80,443,3306
7	Possible network service discovery	172.18.0.31	T1046 - Network Service Discovery	Discovery	Medium	Resolved from STIX	2026-11-01T13:22:59 INFO Network scan detected from 172.18.0.31 against ports 22,80,443,3306
8	Possible network	172.18.0.31	T1046 - Network	Discovery	Medium	Resolved	2026-11-01T13:23:29 INFO Network scan detected from 172.18.0.31 against ports

Figure 8. Dashboard with the simulator running: 29 findings (High 4, Medium 25).

The simulator was then stopped, the original six-line log was restored from the backup (count = 6, git status showed no modified tracked file) and the stack was shut down with docker compose --profile simulation down.

6. SOC narrative

On the clean baseline log, the detector reported two candidate findings. Observed facts: the log contains three failed-password lines from 172.18.0.5 and one line stating that a network scan was detected from 172.18.0.7. Two rules matched: at least three failures from one IP, and the text "network scan detected". Interpretation: the failures are consistent with a possible brute-force attempt (T1110), and the scan line with possible service discovery (T1046). The log does not show whether any login succeeded, who produced the scan message, or whether the activity was authorized testing. A Tier 1 analyst would triage both alerts and escalate the 172.18.0.5 finding; Tier 2 would correlate authentication and network data before concluding malicious intent. The experiments also showed that spreading attempts across IP addresses evades the rule.